Daneben! — Step-by-Step Device Operational Flowchart

Detailed Pipeline Mapping Every Operational State to its Specific Hardware Components and Logic Functions

PHASE 1

Low-Power Presence Ping

Continuous · Every 500ms

HARDWARE COMPONENT

JSN-SR04T External Ultrasonic

Waterproof sealed transducer pointing downward at pavement beside the bin

SYSTEM FUNCTIONALITY

Distance Echo Measurement

Emits 40kHz ultrasound burst (25ms @ 35mA). ESP32 stays in light sleep between pings. Baseline draw: ~3.4mA.

DECISION LOGIC

Object < 1.5m?

- NO: Sleep & repeat
- YES: Trigger Phase 2

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PHASE 2

Pedestrian Filter (Debounce)

Duration: 8.0 Seconds

HARDWARE COMPONENT

ESP32-S3 Hardware RTC Timer

Camera, AI, and Wi-Fi stay completely unpowered via hardware MOSFETs

SYSTEM FUNCTIONALITY

Stationary Object Persistence Check

Times object presence. Passersby walking past clear within 2–3s → Discard event immediately. Zero photos taken. GDPR 100% safe.

DECISION LOGIC

Still present after 8s?

- NO: Return to Phase 1
- YES: Wake Camera (Phase 3)

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PHASE 3

Ground-Level Image Capture

Duration: ~100 Milliseconds

HARDWARE COMPONENT

OV2640 Camera + 8MB PSRAM

Borosilicate optical window, angled 45° down. Physical view frames pavement only (≤0.8m).

SYSTEM FUNCTIONALITY

DMA Framebuffer Transfer

Powers up camera rail via MOSFET. Captures single 96×96 or 160×160 RGB frame. Directly streams raw bytes into PSRAM.

NEXT ACTION

Frame stored in PSRAM Tensor Arena → Pass to Phase 4

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PHASE 4

On-Device TinyML Inference

Duration: ~220 Milliseconds

HARDWARE COMPONENT

ESP32-S3 Vector SIMD Engine

Dual-core 240MHz running ESP-NN hardware-accelerated 128-bit vector instructions

SYSTEM FUNCTIONALITY

Edge Impulse Neural Network

Runs quantized INT8 MobileNetV2/FOMO model (~350KB). Classifies image into: Empty Ground vs Trash Bag vs Cardboard Box.

DECISION LOGIC

Trash Confirmed (>80%)?

- NO: False alarm → Sleep
- YES: Trigger Phase 5

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PHASE 5

Internal Bin Capacity Check

Duration: ~50 Milliseconds

HARDWARE COMPONENT

JSN-SR04T #2 Internal Lid Sensor

Mounted on underside of bin lid pointing down into the waste container

SYSTEM FUNCTIONALITY

Fill-Level Depth Measurement

Fires ultrasound pulse inside the bin. Measures distance to top of garbage. Distance < 20cm means container is completely full.

DECISION LOGIC

Space available inside?

- YES: Trigger Voice (Phase 6)
- NO: Skip voice → Go to Phase 7

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PHASE 6

Interactive Voice Deterrent

Duration: 3.0 Seconds (Audio)

HARDWARE COMPONENT

MAX98357A I2S Amp + 2W Speaker

Acoustic vents protected by waterproof ePTFE hydrophobic membrane (Gore-Tex)

SYSTEM FUNCTIONALITY

Spoken Prompt & Compliance Audit

Plays WAV prompt: "Bitte den Müll in die Tonne werfen. Es ist noch Platz!". Waits 5 mins: logs whether trash was picked up or ignored.

NEXT ACTION

Log status flag: "Warning Played / Ignored" → Go to Phase 7

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PHASE 7

Cloud Alert & Fleet Telemetry

Duration: ~1.0 Second

HARDWARE COMPONENT

ESP32-S3 Wi-Fi Radio (or LoRaWAN)

Radio stays OFF 99.9% of time. Powers on ONLY during verified state change.

SYSTEM FUNCTIONALITY

MQTT Telemetry Packet Broadcast

Transmits JSON: {"bin_id": 402, "illegal_dump": true, "bin_full": false, "warning_ignored": true, "batt_v": 3.84}. Reroutes city trucks.

CYCLE COMPLETION

Radio shuts down → Cut camera power → Enter Light Sleep (Phase 1)